SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

COMPUTER SCIENCE AND ENGINEERING PROGRAMME

CSA37- SOFTWARE TESTING

EXP:11

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Requirement Analysis and Positive Test Scenarios and Test Cases for a Railway Reservation System

Aim

To study the requirements of a Railway Reservation System, and to prepare positive test scenarios and test cases that confirm the application works correctly for valid inputs and normal user actions.

Problem Statement

A railway reservation system lets passengers search trains, check seat availability, book tickets, pay online and cancel tickets. Many users book at the same time, and bookings involve money and limited seats, so a defect can cause wrong bookings, double bookings or refund problems.

Testing is needed to check that the application:

- Passengers can search trains, book tickets and manage bookings correctly.
- Invalid inputs and rule violations are rejected safely.
- Booking limits, payment and cancellation rules follow the requirements.

Any behaviour that differs from the specification must be recorded, with enough detail for developers and testers to repeat and fix it.

Software Requirements

1 Functional requirements (system specification)

ID	Requirement
FR1	The user registers and logs in with a unique user ID and password.
FR2	The user can search trains by source station, destination station and journey date.
FR3	Search results show the train name, number, timings, classes, seat availability and fare.
FR4	A ticket can have up to 6 passengers. Each passenger needs a name, age, gender and an optional berth preference.
FR5	Passenger age must be between 1 and 120, and the name must contain letters only.
FR6	Booking is allowed up to 120 days before the journey date. Past dates are not allowed.
FR7	Payment is possible by UPI, card or net banking within 10 minutes of starting the payment.
FR8	After successful payment, a PNR is generated and the e-ticket is sent by SMS and email.
FR9	When seats are not available, the system books a waiting-list (WL) ticket.
FR10	The user can check the status of a ticket using a 10-digit PNR.
FR11	A ticket can be cancelled before the chart is prepared. The refund follows the cancellation rules.
FR12	Tatkal booking opens at 10:00 AM for AC classes, one day before the journey.

2 Software and environment requirements

Item	Details
Application	Web portal (Java Spring Boot back end, Angular front end)
Database	Oracle / MySQL (users, trains, schedules, bookings, payments)
Operating system	Windows 10/11 or Linux
Browsers	Chrome, Firefox, Edge (latest versions)
Testing tools	Test case document (Excel), Selenium, JMeter, bug tracker (Jira)
Test data	Station codes (MAS, SBC, NDLS), train numbers, passenger details, journey dates, sandbox payment details

Positive Test Scenarios

A positive test scenario checks that a feature works as expected when the user gives valid input and follows the normal flow.

Scenario ID	Module	Test Scenario	Requirement
RR-PS01	Login	Log in with valid credentials	FR1
RR-PS02	Search	Search trains between two stations	FR2, FR3
RR-PS03	Availability	Check availability and fare for a class	FR3
RR-PS04	Booking	Book a ticket for one passenger	FR4, FR5
RR-PS05	Booking	Book a ticket for several passengers with a berth preference	FR4
RR-PS06	Booking window	Book exactly 120 days in advance	FR6
RR-PS07	Payment	Pay using UPI	FR7
RR-PS08	Ticket	PNR and e-ticket after payment	FR8
RR-PS09	Waiting list	Book a ticket when no seats are available	FR9
RR-PS10	PNR status	Check the status of a ticket	FR10
RR-PS11	Cancellation	Cancel a confirmed ticket	FR11
RR-PS12	Tatkal	Book a Tatkal ticket after opening time	FR12

Test Cases

Test Case ID	Scenario ID	Test Steps	Test Data	Expected Result
RR-PT01	RR-PS01	1. Open the login page 2. Enter user ID and password 3. Click Login	Balaguru01; Rail@2026	User is logged in and the booking page is displayed.

Test Case ID	Scenario ID	Test Steps	Test Data	Expected Result
RR-PT02	RR-PS02	1. Enter source and destination 2. Select a valid future date 3. Click Find Trains	MAS to SBC; date 7 days ahead	List of trains is shown with number, timings, classes, availability and fare.
RR-PT03	RR-PS03	1. Open a train from the results 2. Select 3A 3. Click Check Availability	Train 12607; class 3A	Seat availability and the fare are shown for each date.
RR-PT04	RR-PS04	1. Select the train and class 2. Enter passenger details 3. Click Continue	Anand R; age 28; Male	Booking moves to the payment page with the correct fare.
RR-PT05	RR-PS05	1. Add 4 passengers 2. Select Lower berth preference 3. Continue	4 passengers; Lower berth	Details of all passengers are accepted, and the fare is 4 × single fare.
RR-PT06	RR-PS06	1. Select a journey date exactly 120 days from today 2. Search and book	Date = today + 120 days	The booking is allowed at the boundary value.
RR-PT07	RR-PS07	1. Choose UPI 2. Enter a test UPI ID 3. Approve within 10 minutes	success@upi	Payment succeeds and the booking is confirmed.
RR-PT08	RR-PS08	1. Complete the payment 2. Check the confirmation page, SMS and email	Successful booking	A PNR is shown, and the e-ticket is received by SMS and email.
RR-PT09	RR-PS09	1. Select a train and class with no seats 2. Enter passenger details 3. Pay	Class showing WL 5	Ticket is booked with status WL and a waiting-list number.
RR-PT10	RR-PS10	1. Open PNR Status 2. Enter the PNR 3. Click Check	10-digit PNR from a booked ticket	Current status, coach and berth details are displayed.
RR-PT11	RR-PS11	1. Open Booked Tickets 2. Select the ticket 3. Click Cancel 4. Confirm	Confirmed ticket, chart not prepared	Ticket is cancelled and the refund is calculated as per the rules.
RR-PT12	RR-PS12	1. Search for a train for tomorrow 2. At 10:00 AM select an AC class 3. Choose Tatkal quota	Journey date = tomorrow; AC class	Tatkal booking is allowed and the Tatkal fare is applied.

Result / Conclusion

Result: 12 positive test scenarios and 12 test cases were prepared for the Railway Reservation System, covering 12 of the 12 functional requirements (FR1 to FR12). The test cases by module are:

Module	Test Scenarios	Test Cases
Login	1	1
Search	1	1
Availability	1	1
Booking	2	2
Booking window	1	1
Payment	1	1
Ticket	1	1
Waiting list	1	1
PNR status	1	1
Cancellation	1	1
Tatkal	1	1
Total	12	12

Conclusion: The positive test cases cover the main user flows of the Railway Reservation System, from the first action to the final result. When they are executed on a build, they confirm that the valid-input behaviour matches the specification. They should be run first in every test cycle and kept in the regression suite, with the negative test cases added to check invalid inputs.